

02456 Deep Learning

Exercises Week 5

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Exam 2025 Q10 Consider constructing an LSTM for classifying sequences into three classes (a many-to-one mapping). Assume we have one-hot encoded an input sentence to a tensor of shape $[T, V]$, where T is the sequence length and V is the vocabulary size (there is no batch dimension). We want a PyTorch LSTM model that, given such an input, outputs unnormalised logits of shape $[3]$ suitable for use with `nn.CrossEntropyLoss` together with a target tensor containing a single class index of shape $[1]$. Which one of the following code snippets defines this model correctly?

- A.

```
class MyLSTM(nn.Module):
    def __init__(self, V):
        super().__init__()
        self.lstm = nn.LSTM(input_size=V, hidden_size=32)
        self.linear = nn.Linear(32, 3)

    def forward(self, x):
        x, (h_n, c_n) = self.lstm(x)
        return self.linear(x[-1])
```
- B.

```
class MyLSTM(nn.Module):
    def __init__(self, V):
        super().__init__()
        self.lstm = nn.LSTM(input_size=V, hidden_size=32)
        self.linear = nn.Linear(32, 3)

    def forward(self, x):
        x, (h_n, c_n) = self.lstm(x)
        return self.linear(x[: -1])
```
- C.

```
class MyLSTM(nn.Module):
    def __init__(self, V):
        super().__init__()
        self.lstm = nn.LSTM(input_size=V, hidden_size=32)[-1]
        self.linear = nn.Linear(32, 3)

    def forward(self, x):
        x, (h_n, c_n) = self.lstm(x)
        return self.linear(x)
```

```

D.      class MyLSTM(nn.Module):
          def __init__(self, V):
              super().__init__()
              self.lstm = nn.LSTM(input_size=V, hidden_size=32)

          def forward(self, x):
              x, (h_n, c_n) = self.lstm(x)
              return x[-1]

```

A is correct. `x` is the output from our LSTM, and contains all the time steps. Thus an input `[T, V]` results in a tensor `[T, 32]`. The linear layer then expects an input `[32]` to project to `[3]`. As you select the last input of a list in Python with the syntax `x[-1]`, this is the right way to forward pass. Only A passes this through the linear layer.